

Task 3: Microservices with Hazelcast Distributed Map

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перевірити GitHub: <https://github.com/mariiaivanchenko/Software-Architecture/tree/micro-hazelcast>

У цій практичній фактично доповнюється функціонал першої практичної.

До **logging-service** необхідно додати:

- у якості сховища повідомлень **Hazelcast Distributed Map**
- можливість запускати одночасно декілька копій **logging-service**
- **facade-service** випадковим чином обирає до якої копії **logging-service**

звертатись для запису та читання повідомлень

Базова функціональність системи

У якості клієнта цього разу я використовувала файл *requests.http*, де за допомогою розширення *REST Client*, можна легко надсилати запити.

```
requests.http > ...
2 references
1 @FACADE_URL = http://127.0.0.1:5000/facade-service
2
3 ### POST request for facade-service
4 Send Request
5 POST {{FACADE_URL}}
6 Content-Type: text/plain
7
8 msg08
9
10 ### GET request for facade-service
11 Send Request
12 GET {{FACADE_URL}}
```

У завданні необхідно запустити 3 **logging-service** кожен з яких містить **Hazelcast Distributed Map**, приєднати до них відповідні елементи з **Hazelcast** кластеру. Надіслати 10 повідомлень на **facade-service**.

Для запуску усіх необхідних елементів:

- 1 *facade-service*, 1 *messages-service*, 1 *config-server*
- 3 *logging-service*, 3 *Hazelcast cluster nodes*

використовувала скрипт [start.sh](#), який також витягує змінні з *configuration.ini*, відповідний скрипт вбиває усі процеси *stop.sh*.

Запуск елементів

```
.venv➤ micro-hazelcast bash start.sh
Loading configuration file data...
Starting hazelcast nodes...
#####
#####
```

```
Members (size:3, ver:3) [
  Member [127.0.0.1]:5701 - 193365f9-154a-4614-a9b2-5a90559ceaf9
  Member [127.0.0.1]:5703 - 2653ad4d-1fbb-4aeb-be4d-13572cf93472
  Member [127.0.0.1]:5702 - 45f27736-abec-4689-acd6-69273fd78a57 this
]
```

```
Starting services...
Starting logging services and hazelcast nodes...
* Serving Flask app 'facade-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5800
Press CTRL+C to quit
2025-03-10 16:48:57,557 [ INFO] [hz.competent_kirch.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]:
0a [hello-world] [5.5.0] Received auth from ConnectionId=5, /127.0.0.1:5701->/127.0.0.1:58848, qualifier=null, endpoint=[127.0.0.1]:
1f59217c-4771-4f2d-9aeb-67da17eda18, alive=true, connectionType=PM, planeIndex=-1, successfully authenticated, clientId=
71-4f2d-9aeb-67da17eda18, client name: hz.client_0, client version: 5.5.0
2025-03-10 16:48:57,798 [ INFO] [hz.cool_oparin.generic-operation.thread-2] [c.h.c.i.p.t.AuthenticationMessageTask]: [127.0.0.1]:
r10] [5.5.0] Received auth from ConnectionId=5, /127.0.0.1:5703->/127.0.0.1:58841, qualifier=null, endpoint=[127.0.0.1]:58841, r
17c-4771-4f2d-9aeb-67da17eda18, alive=true, connectionType=PM, planeIndex=-1, successfully authenticated, clientId: 1f59217c
-4fda17eda18, client name: hz.client_0, client version: 5.5.0
2025-03-10 16:48:57,888 [ INFO] [hz.heuristic_meninsky.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTas
:5702] [hello-world] [5.5.0] Received auth from ConnectionId=5, /127.0.0.1:5702->/127.0.0.1:58842, qualifier=null, endpoint=[127.
matchId:1f59217c-4771-4f2d-9aeb-67da17eda18, alive=true, connectionType=PM, planeIndex=-1, successfully authenticated, client
-4771-4f2d-9aeb-67da17eda18, client name: hz.client_0, client version: 5.5.0
* Serving Flask app 'logging-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5801
Press CTRL+C to quit
```

```
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5003
Press CTRL+C to quit
* Serving Flask app 'messages-service'
* Debug mode: off
* Serving Flask app 'config-server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5004
Press CTRL+C to quit
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5005
Press CTRL+C to quit
```

Надсилаємо повідомлення

спершу **GET**, перевірити, що повідомлень ще не надходило.

```
Response(1118ms) X
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Mon, 10 Mar 2025 14:43:56 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 75
6 Connection: close
7
8 Logging response:
9 Messages response: Messages GET is not implemented yet.
10
```

Коли надходять запити на **facade-service**:

- якщо це **POST**, надсилаємо запит на **config-server**, щоб отримати перелік можливих **logging-service** для надсилання запиту.
після того рандомно вибираємо до якого з них звернутись, при тому перевіряємо чи можна до нього виконати запит, виводимо вибрану адресу.
- якщо це **GET**, також робимо запит на **config-server**, але тепер ще й для адрес для **messages-service**. після так само рандомно вибираємо, яку доступну адресу викликати, виводимо вибрану адресу.

msg1 - <http://127.0.0.1:5002/logging-service>

```
127.0.0.1 - - [10/Mar/2025 16:44:54] "GET /config-server?service_name=logging HTTP/1.1" 200 -  
127.0.0.1 - - [10/Mar/2025 16:44:54] "GET /logging-service HTTP/1.1" 200 -  
Chooesd logging in POST: http://127.0.0.1:5002/logging-service  
Recieved messages: msg1  
127.0.0.1 - - [10/Mar/2025 16:44:55] "POST /logging-service HTTP/1.1" 200 -  
127.0.0.1 - - [10/Mar/2025 16:44:55] "POST /facade-service HTTP/1.1" 200 -
```

msg2 - <http://127.0.0.1:5001/logging-service>

```
127.0.0.1 - - [10/Mar/2025 16:52:50] "GET /config-server?service_name=logging HTTP/1.1" 200 -  
127.0.0.1 - - [10/Mar/2025 16:52:50] "GET /logging-service HTTP/1.1" 200 -  
Chooesd logging in POST: http://127.0.0.1:5001/logging-service  
Recieved messages: msg2  
127.0.0.1 - - [10/Mar/2025 16:52:50] "POST /logging-service HTTP/1.1" 200 -  
127.0.0.1 - - [10/Mar/2025 16:52:50] "POST /facade-service HTTP/1.1" 200 -
```

msg3 - <http://127.0.0.1:5003/logging-service>

```
127.0.0.1 - - [10/Mar/2025 16:53:12] "GET /config-server?service_name=logging HTTP/1.1" 200 -  
127.0.0.1 - - [10/Mar/2025 16:53:12] "GET /logging-service HTTP/1.1" 200 -  
Chooesd logging in POST: http://127.0.0.1:5003/logging-service  
Recieved messages: msg3  
127.0.0.1 - - [10/Mar/2025 16:53:13] "POST /logging-service HTTP/1.1" 200 -  
127.0.0.1 - - [10/Mar/2025 16:53:13] "POST /facade-service HTTP/1.1" 200 -
```

msg4 - <http://127.0.0.1:5001/logging-service>

```
127.0.0.1 -- [10/Mar/2025 16:53:48] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:53:49] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5001/logging-service
Recieved messages: msg4
127.0.0.1 -- [10/Mar/2025 16:53:49] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:53:49] "POST /facade-service HTTP/1.1" 200 -
```

msg5 - <http://127.0.0.1:5001/logging-service>

```
127.0.0.1 -- [10/Mar/2025 16:54:20] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:54:20] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5001/logging-service
Recieved messages: msg5
127.0.0.1 -- [10/Mar/2025 16:54:20] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:54:20] "POST /facade-service HTTP/1.1" 200 -
```

msg6 - <http://127.0.0.1:5003/logging-service>

```
127.0.0.1 -- [10/Mar/2025 16:54:39] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:54:39] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5003/logging-service
Recieved messages: msg6
127.0.0.1 -- [10/Mar/2025 16:54:39] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:54:39] "POST /facade-service HTTP/1.1" 200 -
```

msg7 - <http://127.0.0.1:5002/logging-service>

```
127.0.0.1 -- [10/Mar/2025 16:54:59] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:54:59] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5002/logging-service
Recieved messages: msg7
127.0.0.1 -- [10/Mar/2025 16:54:59] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:54:59] "POST /facade-service HTTP/1.1" 200 -
```

msg8 - <http://127.0.0.1:5003/logging-service>

```
127.0.0.1 -- [10/Mar/2025 16:55:25] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:55:25] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5003/logging-service
Recieved messages: msg8
127.0.0.1 -- [10/Mar/2025 16:55:25] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 16:55:25] "POST /facade-service HTTP/1.1" 200 -
```

msg9 - <http://127.0.0.1:5001/logging-service>

```
127.0.0.1 - - [10/Mar/2025 16:55:49] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:55:49] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5001/logging-service
Recieved messages: msg9
127.0.0.1 - - [10/Mar/2025 16:55:49] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:55:49] "POST /facade-service HTTP/1.1" 200 -
```

msg10 - <http://127.0.0.1:5002/logging-service>

```
127.0.0.1 - - [10/Mar/2025 16:56:10] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:56:10] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in POST: http://127.0.0.1:5002/logging-service
Recieved messages: msg10
127.0.0.1 - - [10/Mar/2025 16:56:10] "POST /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:56:10] "POST /facade-service HTTP/1.1" 200 -
```

знову робимо **GET**.

```
Response(124ms) ×
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Mon, 10 Mar 2025 14:57:05 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 134
6 Connection: close
7
8 Logging response: msg2, msg1, msg6, msg5, msg4, msg7, msg8, msg9, msg3, msg10
9 Messages response: Messages GET is not implemented yet.
10
```

```
127.0.0.1 - - [10/Mar/2025 16:57:05] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:57:05] "GET /config-server?service_name=messages HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:57:05] "GET /logging-service HTTP/1.1" 200 -
Chooesd logging in GET: http://127.0.0.1:5002/logging-service
127.0.0.1 - - [10/Mar/2025 16:57:05] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:57:05] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:57:05] "GET /facade-service HTTP/1.1" 200 -
```

якщо зробити ще один запит, але на інший елемент кластеру, повідомлення будуть однакові.

```
Response(137ms) X
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Mon, 10 Mar 2025 14:58:18 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 134
6 Connection: close
7
8 Logging response: msg2, msg1, msg6, msg5, msg4, msg7, msg8, msg9, msg3, msg10
9 Messages response: Messages GET is not implemented yet.
10
```

```
127.0.0.1 - - [10/Mar/2025 16:58:18] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:58:18] "GET /config-server?service_name=messages HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:58:18] "GET /logging-service HTTP/1.1" 200 -
Chosed logging in GET: http://127.0.0.1:5001/logging-service
127.0.0.1 - - [10/Mar/2025 16:58:18] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:58:18] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 16:58:18] "GET /facade-service HTTP/1.1" 200 -
```

також, трошки змінило логіку deduplication в порівнянні з попередньою роботою. тепер повідомлення мають відрізнятись і контентом, і uuid кодом.

Вимкнути елементи кластера

вибираю два елементи та вбиваю їх.

```
ps aux | grep flask
kill -9 44208 44092
```

```
● .venv+ micro-hazelcast ps aux | grep flask
maria 44208 1.4 0.1 33672680 6312 s031 S+ 4:40PM 0:16.09 /usr/local/Cellar/python@3.13/3.13.2/Frameworks/Python.framework/Versions/3.13/Resources/Python.app/Contents/MacOS/Python /Users/mariia/Desktop/Task3/micro-hazelcast/.venv/bin/flask --app logging-service run --port=5002
maria 44092 1.4 0.1 33672672 6428 s031 S+ 4:40PM 0:16.29 /usr/local/Cellar/python@3.13/3.13.2/Frameworks/Python.framework/Versions/3.13/Resources/Python.app/Contents/MacOS/Python /Users/mariia/Desktop/Task3/micro-hazelcast/.venv/bin/flask --app logging-service run --port=5001
maria 44223 1.4 0.1 33673696 6300 s031 S+ 4:41PM 0:15.87 /usr/local/Cellar/python@3.13/3.13.2/Frameworks/Python.framework/Versions/3.13/Resources/Python.app/Contents/MacOS/Python /Users/mariia/Desktop/Task3/micro-hazelcast/.venv/bin/flask --app logging-service run --port=5003
maria 60340 0.0 0.0 33610048 788 s033 S+ 5:02PM 0:00.01 grep --color=auto --exclude-dir=.bzl --exclude-dir=CVS --exclude-dir=.git --exclude-dir=.hg --exclude-dir=.svn --exclude-dir=.idea --exclude-dir=.tox --exclude-dir=.venv --exclude-dir=.venv flask
maria 44236 0.0 0.0 33652140 4052 s031 S+ 4:41PM 0:00.69 /usr/local/Cellar/python@3.13/3.13.2/Frameworks/Python.framework/Versions/3.13/Resources/Python.app/Contents/MacOS/Python /Users/mariia/Desktop/Task3/micro-hazelcast/.venv/bin/flask --app config-server run --port=5005
maria 44235 0.0 0.0 33651116 4052 s031 S+ 4:41PM 0:00.56 /usr/local/Cellar/python@3.13/3.13.2/Frameworks/Python.framework/Versions/3.13/Resources/Python.app/Contents/MacOS/Python /Users/mariia/Desktop/Task3/micro-hazelcast/.venv/bin/flask --app messages-service run --port=5004
maria 44090 0.0 0.1 33719932 4772 s031 S+ 4:40PM 0:01.39 /usr/local/Cellar/python@3.13/3.13.2/Frameworks/Python.framework/Versions/3.13/Resources/Python.app/Contents/MacOS/Python /Users/mariia/Desktop/Task3/micro-hazelcast/.venv/bin/flask --app facade-service run --port=5000
● .venv+ micro-hazelcast kill -9 44208 44092
○ .venv+ micro-hazelcast
```

після того надсилає запит **GET**, де видно, що намагалось отримати доступ до двох вимкнених елементів, тому обрало третій, що лишився.

```
127.0.0.1 -- [10/Mar/2025 17:03:28] "GET /config-server?service_name=logging HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 17:03:28] "GET /config-server?service_name=messages HTTP/1.1" 200 -
Service at http://127.0.0.1:5002/logging-service is down. Error: HTTPConnectionPool(host='127.0.0.1', port=5002): Max retries exceeded with url: /logging-service (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10d1c39b0>: Failed to establish a new connection : [Errno 61] Connection refused')). Trying another service.
Service at http://127.0.0.1:5002/logging-service is down. Error: HTTPConnectionPool(host='127.0.0.1', port=5002): Max retries exceeded with url: /logging-service (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10d1c3240>: Failed to establish a new connection : [Errno 61] Connection refused')). Trying another service.
Service at http://127.0.0.1:5002/logging-service is down. Error: HTTPConnectionPool(host='127.0.0.1', port=5002): Max retries exceeded with url: /logging-service (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10d1c3460>: Failed to establish a new connection : [Errno 61] Connection refused')). Trying another service.
Service at http://127.0.0.1:5001/logging-service is down. Error: HTTPConnectionPool(host='127.0.0.1', port=5001): Max retries exceeded with url: /logging-service (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x10d1c3790>: Failed to establish a new connection : [Errno 61] Connection refused')). Trying another service.
127.0.0.1 -- [10/Mar/2025 17:03:28] "GET /logging-service HTTP/1.1" 200 -
Chooed logging in GET: http://127.0.0.1:5003/logging-service
127.0.0.1 -- [10/Mar/2025 17:03:28] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 17:03:28] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 -- [10/Mar/2025 17:03:28] "GET /facade-service HTTP/1.1" 200 -
```

результат такий самий як і до.

```
Response(129ms) ×
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Mon, 10 Mar 2025 15:03:28 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 134
6 Connection: close
7
8 Logging response: msg2, msg1, msg6, msg5, msg4, msg7, msg8, msg9, msg3, msg10
9 Messages response: Messages GET is not implemented yet.
10
```

Вбиваємо всі елементи

решту запущених елементів системи

```
⊗ .venv → micro-hazelcast bash stop.sh
Stopping Hazelcast nodes...
Stopping services...
○ .venv → micro-hazelcast █
```

```

Chooed logging in GET: http://127.0.0.1:5003/logging-service
127.0.0.1 - - [10/Mar/2025 17:03:28] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 17:03:28] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 - - [10/Mar/2025 17:03:28] "GET /facade-service HTTP/1.1" 200 -
start.sh: line 63: 43904 Killed: 9          $HAZELCAST_BIN_PATH/hz start -c $HAZELCAST_XML_PATH
start.sh: line 63: 44090 Killed: 9          flask --app facade-service run --port=$PORTS_FACADE_PORT
start.sh: line 63: 44092 Killed: 9          flask --app logging-service run --port=$port
start.sh: line 63: 44208 Killed: 9          flask --app logging-service run --port=$port
start.sh: line 63: 44223 Killed: 9          flask --app logging-service run --port=$port
start.sh: line 63: 44235 Killed: 9          flask --app messages-service run --port=$PORTS_MESSAGES_PORT
start.sh: line 63: 44236 Killed: 9          flask --app config-server run --port=$PORTS_CONFIG_PORT
venv: micro-hazelcast

```

Додатковий функціонал системи

для додаткового функціоналу треба було додати підтримку **config-server**, який би витягував усі необхідні IP адреси сервісів.

відповідна підтримка всередині методів **facade-service**.

```

@app.route("/facade-service", methods=["GET"])
def get():
    """
    Handling GET request from client + Retry logic.
    """
    result_message = ""
    retry_attempts = 3

    config_url_logging = CONFIG_URL + "?service_name=logging"
    config_url_messages = CONFIG_URL + "?service_name=messages"

    logging_addresses_response = requests.get(config_url_logging, timeout=3)
    messages_addresses_response = requests.get(config_url_messages, timeout=3)

    logging_addresses = logging_addresses_response.text.split(" ")
    messages_addresses = messages_addresses_response.text.strip()

    logging_url = choose_logging(logging_addresses)
    messages_url = messages_addresses
    print("Chooed logging in GET: ", logging_url)

```

```

72 @app.route("/facade-service", methods=["POST"])
73 def post():
74     """
75     Handling POST request from client + Retry logic.
76     """
77     retry_attempts = 3
78     message = request.get_data(as_text=True)
79     random_uuid = uuid.uuid4()
80     data = {"uuid": str(random_uuid), "message": message}
81     headers = {"Content-type": "application/json"}
82
83     config_url_logging = CONFIG_URL + "?service_name=logging"
84     logging_addresses_response = requests.get(config_url_logging, timeout=3)
85     logging_addresses = logging_addresses_response.text.split(" ")
86     logging_url = choose_logging(logging_addresses)
87     print("Chooed logging in POST: ", logging_url)

```

config-server.py

```

import configparser
from flask import Flask, request

app = Flask(__name__)

config = configparser.ConfigParser()
config.read("configuration.ini")

```

```

@app.route("/config-server", methods=["GET"])
def get():
    service_name = request.args.get("service_name", default="Guest", type=str)
    if service_name == "logging":
        logging_urls = [
            member.strip().strip('"')
            for member in config.get("URLS", "LOGGING_URLS").split(" , ")
        ]
        return " , ".join(logging_urls)

    if service_name == "messages":
        messages_url = config.get("URLS", "MESSAGES_URL")
        return messages_url

    return "Invalid service name", 400

if __name__ == "__main__":
    app.run(debug=True)

```